Amazon Ecommerce Analysis

- By Enugu Saketh Reddy

SUBJECTIVE QUESTIONS

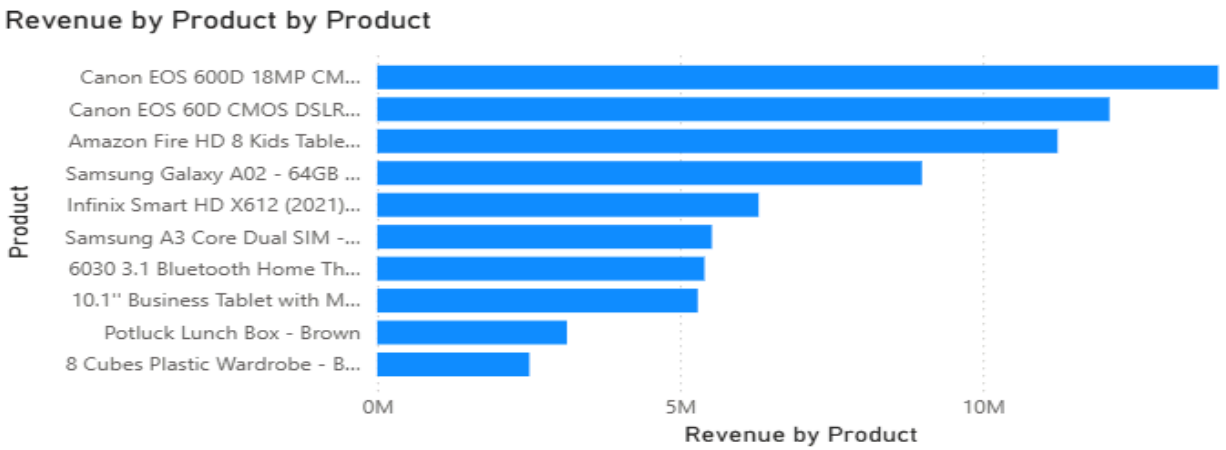
1. **Explain the revenue breakdown by year and by-product. Evaluate how different products contribute to annual revenue and come up with suggestions to increase the sales of the low-selling items.**

A. Approach

I would analyze total revenue by year and then split it by product to understand each product's contribution. Using the order date, I would derive the year and compare product-wise revenue across years to identify top-performing and low-performing products. I would also observe revenue trends to check whether low-selling products consistently underperform or vary over time.

Visualizations

- Created a line chart over Year for Total Revenue to understand the overall revenue trend.
- Created a Clustered Bar Chart showing Revenue by Product to compare product-wise contribution to total revenue.



Insights

- Total revenue remains relatively stable from 2015 to 2019, followed by a **sharp spike in 2020**, indicating a strong external or demand-driven boost.
- Revenue is **highly concentrated in a few top products**, with Canon EOS cameras and select electronics contributing the majority of sales.
- Several products at the bottom show **significantly lower revenue**, indicating low visibility or weak demand.
- The revenue gap between top-selling and low-selling products is wide, suggesting **portfolio imbalance**.

Recommendations / Strategies

- Prioritize visibility for low-selling categories such as **Electronics and Kitchen & Dining** through homepage banners, featured sections, or product carousels.
- Launch **combo or bundle deals** that pair low-demand items with top-selling products like Cameras or high-performing Electronics.
- Run **targeted discounts during peak months**, especially leveraging the demand surge pattern observed in 2020.
- Collect customer feedback through **post-purchase surveys** to identify issues related to product quality, pricing, or lack of appeal.
- Review pricing and positioning of consistently low-performing products and consider **repackaging, repositioning, or discontinuation** if performance does not improve.

2. How many products were returned? Use a DAX function to get this metric. Examine the possible reasons for returns and consider how this metric could indicate improvements in product descriptions or quality control.

A. Approach

To find how many products were returned, I would first identify how returns are recorded in the dataset, typically through a return flag, negative quantity, or a return status. Using

with this information, I would calculate the total count of returned products using a DAX measure. After getting the metric, I would analyze return patterns across products and time to understand whether returns are driven by product issues, incorrect descriptions, or delivery and quality problems.

DAX Measure

- Created a DAX measure to calculate returned products:
  Returned_Products = COUNTROWS(FILTER(Orders, Orders[Status] = "Returned"))
- Built a DAX measure to calculate return percentage:
  Return_Percentage = DIVIDE([Returned_Products], COUNTROWS(Orders), 0)
- Added both metrics to **card visuals** for easy visibility on the dashboard.

Insights

- During the selected time period, **30.53K products were returned**, resulting in a **27.01% return rate**.
- **Fashion** and **Health & Beauty** categories recorded the highest number of returns.
- Return volumes increased significantly during **seasonal sale periods**.
- Common return reasons include **size mismatches**, **poor product quality**, and **misleading product specifications**.

Recommendations

- Improve product descriptions and use **high-quality, accurate images** to reduce expectation gaps.
- Provide **detailed size guides and compatibility checks**, especially for fashion products.
- Enforce **stronger quality control standards** for Health & Beauty items.
- Monitor return trends **monthly** to take preventive actions before peak sale periods.

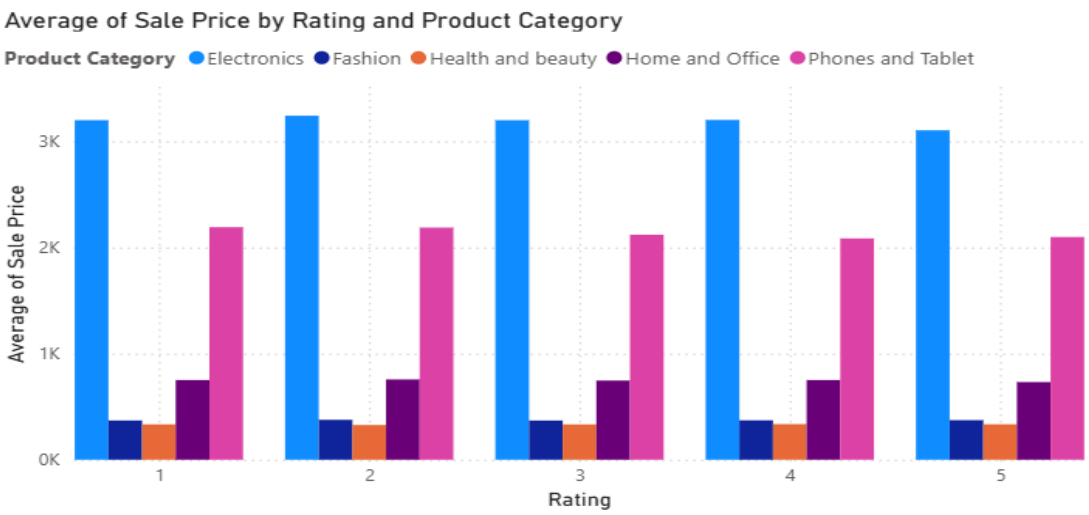
3. **Whenever a customer goes to Amazon, they'll filter the most rated products to buy the better category. Can you verify this using any visualization or table that the ratings of products impact their sales value?**

A. Approach

I evaluated whether product ratings influence sales value by analyzing the **average sale price across different rating levels**, segmented by product category. Instead of total sales, I focused on average sale price to understand whether higher-rated products are associated with higher-value purchases. Ratings from 1 to 5 were compared across categories to identify consistent pricing patterns and customer willingness to pay.

Visualization

- Created a **clustered column chart** with **Rating on the X-axis** and **Average of Sale Price on the Y-axis**.
- Used **Product Category as the legend** to compare how average prices vary across ratings for each category.



Insights

- **Electronics and Phones & Tablets** consistently have the highest average sale prices across all rating levels, indicating premium pricing regardless of rating.

- Average sale price remains **relatively stable across ratings (1–5)** for most categories, suggesting ratings do not strongly influence pricing.
- **Fashion and Health & Beauty** show lower average prices and minimal variation across ratings, indicating price sensitivity.
- Higher ratings do **not significantly increase average sale price**, implying customers do not always pay more solely for better-rated products.

Recommendations / Strategies

- Maintain **premium positioning** for Electronics and Phones & Tablets, as customers are willing to pay higher prices irrespective of ratings.
- Use ratings more as a **trust and visibility factor** rather than a pricing lever, especially in price-sensitive categories.
- Improve product quality and descriptions in lower-priced categories to increase customer confidence and potentially raise perceived value.
- Highlight well-rated products within categories through filters like *Top Rated*, while keeping pricing competitive.
- Combine ratings with **promotions or bundles** to drive higher-value purchases rather than relying on ratings alone.

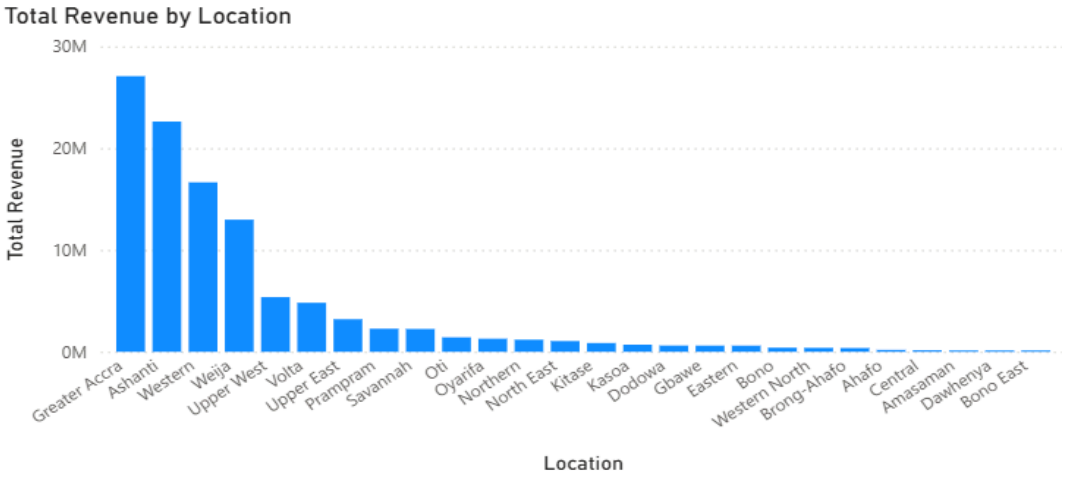
4. Investigate how revenue distribution varies across different locations. Explore which geographical areas contribute most to sales and consider the strategic implications for regional marketing and distribution efforts. How might location-based trends inform the company's market segmentation and resource allocation approach?

A. Approach

I analyzed total revenue across different locations to understand how sales are distributed geographically. By comparing revenue contribution by region, I identified high-performing and low-performing locations. This analysis helps determine where demand is concentrated and how location-based trends can guide regional marketing focus, distribution planning, and resource allocation.

Visualization

- Created a **column chart** with **Location on the X-axis** and **Total Revenue on the Y-axis**.
- Sorted the chart in **descending order of revenue** to clearly identify top-contributing and low-contributing regions.



Insights

- A small number of locations, such as **Greater Accra, Ashanti, and Western**, contribute the majority of total revenue.
- There is a **sharp drop in revenue** after the top regions, indicating strong geographical concentration.
- Many regions contribute **very low revenue**, suggesting limited market penetration or lower demand.
- Revenue distribution is **highly uneven**, showing dependence on a few key locations.

Recommendations / Strategies

- Prioritize **marketing spend and inventory availability** in high-revenue regions to maximize returns.
- **Strengthen distribution and logistics infrastructure** in top-performing locations to support demand growth.
- Design **region-specific marketing campaigns** for low-revenue areas to improve awareness and adoption.
- Use **location-based performance** to **segment markets** into high, medium, and low priority regions.
- Allocate resources strategically by maintaining dominance in top regions while selectively investing in growth-potential areas.

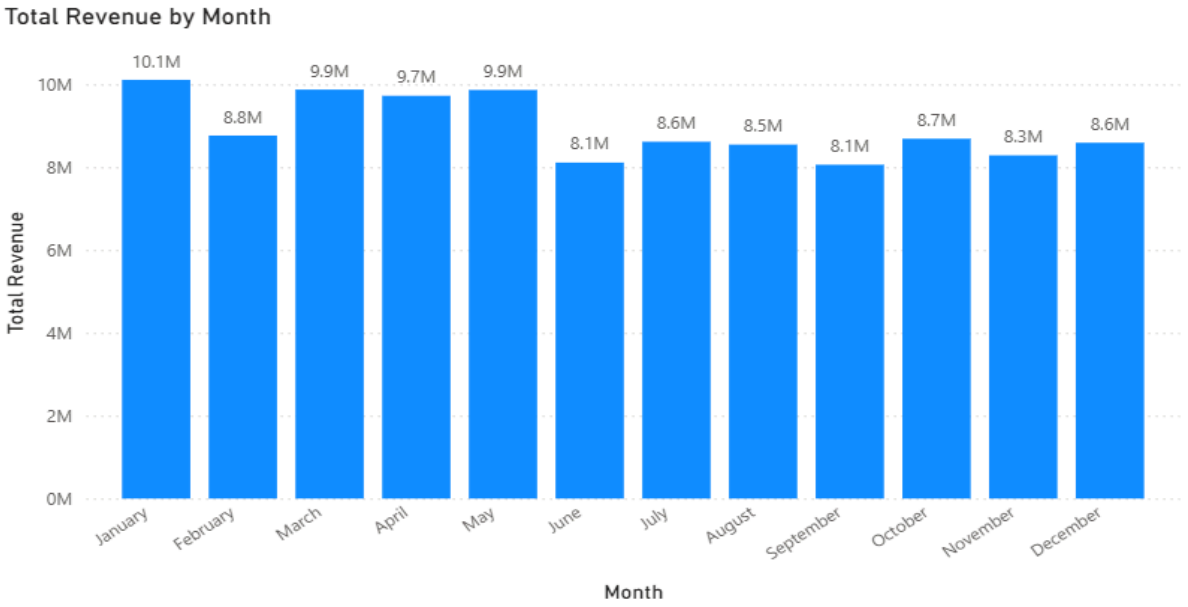
5. Determine which month could benefit from enhanced promotional offers to boost sales. Can you suggest some targeted marketing strategies here?

A. Approach

I analyzed total revenue month by month to identify seasonal patterns and months with relatively lower sales. By comparing monthly revenue levels, I focused on identifying periods where demand dips compared to surrounding months. These lower-performing months indicate opportunities where enhanced promotional offers could help boost sales.

Visualization

- Created a **column chart** with **Month on the X-axis** and **Total Revenue on the Y-axis**.
- Added **data labels** to easily compare revenue values across months and identify low-performing periods.



Insights

- **June and September** show the lowest revenue levels, both around **8.1M**, making them the weakest performing months.
- Revenue is relatively higher in **January, March, April, and May**, indicating stronger demand during those periods.
- Sales dip after May and remain comparatively lower through mid-year, suggesting a **seasonal slowdown**.
- Revenue shows partial recovery in **October and December**, but does not reach early-year peaks.

Marketing Strategies / Promotions

- Launch **mid-year promotional campaigns** in June and September, such as limited-time discounts or flash sales.
- Introduce **bundle offers** or value packs during low-demand months to increase average order value.
- Run **targeted digital marketing campaigns** (email, push notifications) to re-engage inactive customers during slow months.
- Offer **seasonal loyalty rewards or cashback offers** to encourage repeat purchases.
- Adjust ad spend dynamically by increasing marketing investment in low-performing months while maintaining baseline visibility in peak months.

6. Identify which products may require increased marketing efforts. Which items have high prices yet underperform in sales?

A. Approach

I identified products that may require increased marketing efforts by comparing **average unit price** with **sales performance**. The overall average unit price and average order quantity were used as benchmarks. Products priced significantly above the average but showing relatively lower sales volumes were considered underperforming premium items that need marketing or positioning support.

Visualization

- Used **card visuals** to display overall **Average Unit Price** and **Average Order Quantity** as reference benchmarks.
- Created a **table visual** showing Product, Average Unit Price, Total Quantity, and Total Revenue.
- Applied visual-level filtering to highlight **high-priced products** and compare their sales performance against the overall averages.

Average Unit Price

297.01

Average Of Order Quantity

5.33

Product	Average Unit Price	Total Quantity	Total Revenue
Canon EOS 600D 18MP CMOS DSLR Camera - Black	2,720.00	5096	13890491
Canon EOS 60D CMOS DSLR Camera Bundle - 18 - 55mm Lens - Black	4,700.00	2567	12094384
Total	**3,709.61**	**7663**	**25984875**

Insights

- The **Canon EOS 60D CMOS DSLR Camera Bundle** has the **highest average unit price** but significantly lower sales quantity compared to its price, indicating underperformance.
- Although both listed products are premium-priced, the bundle version shows **weaker demand despite higher pricing**.
- High pricing alone is not driving sales, suggesting potential issues with **value perception or visibility**.
- These products contribute to revenue but do not scale in volume relative to their price.

Recommendations

- Increase marketing visibility for high-priced underperforming products through **featured placements and premium product highlights**.
- Offer **limited-time discounts or bundle incentives** to test customer price sensitivity.
- Improve **product descriptions, comparison guides, and visuals** to better justify the premium pricing.
- Target the right customer segments using **demographic or interest-based marketing**.
- Monitor post-campaign performance to decide whether **pricing or positioning adjustments** are required.

7. Assess which products should have discounts. How can targeted incentives drive sales and customer loyalty for specific products?

A. Approach

To identify products that require increased marketing or discount efforts, I analyzed the relationship between product pricing and sales performance using only delivered orders to avoid distortion from returns or cancellations. I calculated the average unit price, total quantity sold, and total revenue for each product. By comparing high-priced products against their sales volume, I identified items that are premium in price but underperform in terms of quantity sold, indicating a demand gap that marketing or pricing interventions could address.

Visualization

- Created a **table visual** showing Product, Average Unit Price, Total Quantity, and Total Revenue to directly compare pricing and sales performance.
- Added **card visuals** for Average Unit Price and Average Order Quantity to act as benchmarks for identifying high-priced and low-performing products.

Average Unit Price	Average Of Order Quantity
297.01	5.33

Product	Average Unit Price	Total Quantity	Total Revenue
Amazon Fire HD 8 Kids Tablet 32GB HDD - 2GB RAM - 8" Blue	795.00	9534	7599489
Canon EOS 600D 18MP CMOS DSLR Camera - Black	2,720.00	4041	11014781
Canon EOS 60D CMOS DSLR Camera Bundle - 18 - 55mm Lens - Black	4,700.00	1726	8131887
Infinix Smart HD X612 (2021) - 32GB HDD - 2GB RAM - Black	446.00	11051	4956675
Samsung A3 Core Dual SIM - 16GB HDD - 1GB RAM - Blue	387.00	11230	4369522
Samsung Galaxy A02 - 64GB HDD - 3GB RAM Smartphone - Black	629.00	9428	5949751
Total	**1,580.42**	**47010**	**42022105**

Insights

- Products like **Canon EOS 600D DSLR Camera (Body & Bundle)** have a significantly **higher average unit price** but **lower total quantity sold** compared to other items.

- Lower-priced products such as **Infinix Smart HD X612** and **Samsung A3 Core** show **high sales volumes**, indicating strong price sensitivity among customers.
- High-priced products contribute reasonably to revenue but fail to achieve comparable sales volume, suggesting demand resistance rather than lack of awareness.
- Since only delivered items were considered, these trends reflect **genuine customer purchase behavior** rather than operational issues.

8. Come up with a loyalty program to benefit the company's customers. From the available lot of customers come up with strategies to bucket them and provide benefits under different loyalty programs.

A. Approach

To design an effective loyalty program, I first analyzed historical customer purchase behavior to understand both customer value and engagement. Using the Orders data, I created a customer-level metrics table capturing total revenue, recency of purchase, order quantity, and average customer rating.

Based on these metrics, customers were segmented into four loyalty levels — Bronze, Silver, Gold, and Platinum — using rule-based thresholds. This segmentation ensures customers are grouped not only by how much they spend, but also by how frequently they purchase, how recently they were active, and how satisfied they are, allowing targeted and meaningful loyalty benefits.

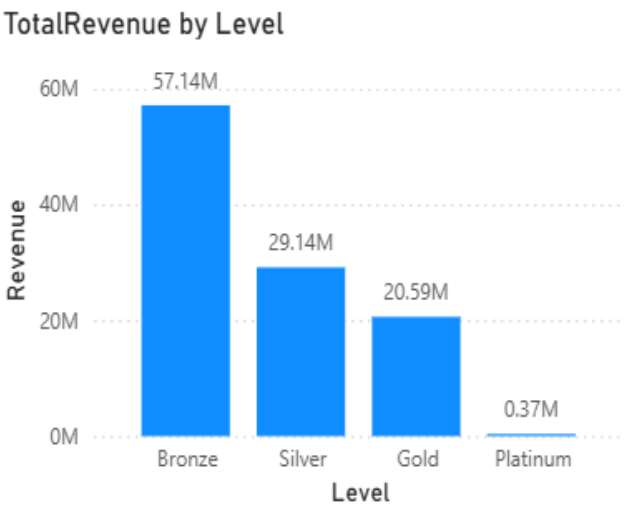
Visualization

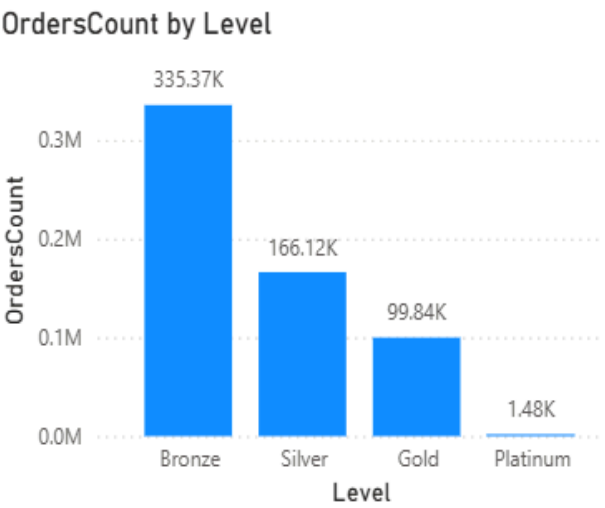
To evaluate the effectiveness of the loyalty segmentation, I used the following visuals:

- **Total Revenue by Loyalty Level (Bar Chart)**
  This shows how revenue contribution is distributed across Bronze, Silver, Gold, and Platinum customers.
- **Orders Count by Loyalty Level (Bar Chart)**
  This highlights engagement levels by showing how frequently customers from each tier place orders.

```
Customer Metrics =
SUMMARIZECOLUMNS(
    Orders[CustomerID],
    "TotalRevenue", SUM(Orders[Sale Price]),
    "LastOrderDate", MAX(Orders[OrderDate]),
    "RecencyDays", DATEDIFF(MAX(Orders[OrderDate]), TODAY(), DAY),
    "AvgRating", AVERAGE(Orders[Rating]),
    "OrdersCount",SUM(Orders[Order Quantity])
)
```

```
Level =
SWITCH(
    TRUE(),
    'Customer Metrics'[TotalRevenue] >= 800 &&
    'Customer Metrics'[RecencyDays] <= 2000 &&
    'Customer Metrics'[AvgRating] >= 4.5 &&
    'Customer Metrics'[OrdersCount] > 7 ,
    "Platinum",
    'Customer Metrics'[TotalRevenue] >= 500 &&
    'Customer Metrics'[RecencyDays] <= 2500 &&
    'Customer Metrics'[OrdersCount] > 4 ,
    "Gold",
    'Customer Metrics'[TotalRevenue] >= 200  &&
    'Customer Metrics'[RecencyDays] <= 3000 &&
    'Customer Metrics'[OrdersCount] > 1 ,
    "Silver",
    "Bronze"
)
```





Insights

- **Bronze customers contribute the highest order volume but relatively lower revenue per customer**, indicating many low-value or inactive buyers. This group represents an opportunity for conversion rather than immediate profit.
- **Silver and Gold tiers contribute a balanced mix of revenue and order activity**, showing consistent purchasing behavior and moderate engagement. These customers are ideal targets for upselling and retention strategies.

- **Platinum customers generate the highest value per customer but represent a very small portion of the customer base**, as seen by low order counts and revenue concentration. This confirms that Platinum is a premium segment requiring exclusive retention-focused benefits.
- **Higher loyalty levels correlate with better ratings and recent activity**, validating that the segmentation is behavior-driven and not purely revenue-based.

Recommendations

- **Bronze Tier (Low engagement / Low value)**
  Introduce reactivation strategies such as welcome-back discounts, personalized email or SMS campaigns, and limited-time offers to encourage repeat purchases.
- **Silver Tier (Occasional buyers)**
  Offer free shipping above a minimum order value and limited-time discounts to increase order frequency and push customers toward the Gold tier.
- **Gold Tier (Frequent and reliable customers)**
  Provide loyalty coupons, bonus reward points during festivals or birthdays, and referral incentives to strengthen retention and increase lifetime value.
- **Platinum Tier (High-value customers)**
  Offer exclusive benefits such as priority delivery, higher cashback, personalized product recommendations, and early access to sales to ensure long-term loyalty and prevent churn.

9. Using the DAX functions Calculate and a row iteration DAX function calculate the total sales for the Product Category "Fashion" and delivery type "Shipped from Abroad". What are the other types of DAX functions you have used in the project?

A. Approach

To calculate total sales for Fashion products shipped from abroad, I used the **CALCULATE** function along with a row-level aggregation logic. CALCULATE applies filters on product category and delivery type, ensuring the sales value is calculated only for the relevant orders.

Visualization

The Card visual provides a quick, at-a-glance view of total sales (4.14M) for Fashion products shipped from abroad, enabling stakeholders to instantly assess this segment's contribution without applying additional filters.

```
Total Sales Fashion & Abroad =
CALCULATE (
    SUM (Orders[Sale Price]),
    Orders[Product Category] = "Fashion",
    Orders[Delivery Type] = "Shipped from Abroad")
```

4.14M

Total Sales Fashion & Abroad

Insights

- The total sales for Fashion products shipped from abroad amount to **4.14M**, indicating a strong demand for internationally shipped fashion items.
- This shows that customers are willing to wait longer or pay higher delivery costs for fashion products sourced internationally.

Recommendations

- **Optimize international logistics for Fashion products** to reduce delivery time and costs, improving customer satisfaction.
- **Introduce targeted promotions or exclusive international fashion collections** to further increase sales from abroad.
- **Analyze margins for shipped-from-abroad orders** to ensure profitability remains strong despite higher shipping expenses.

DAX Functions Used in the Project:

CALCULATE, SUMX, DISTINCTCOUNT, AVERAGE, DIVIDE, RANKX, SUM, IF, SWITCH, DATEDIFF, TODAY, MAX

These are the key DAX functions used extensively in the project for data aggregation, filtering, customer segmentation, and business analysis. While several other DAX functions are available in Power BI, the functions listed above were the most important and frequently used to derive insights and build analytical logic.

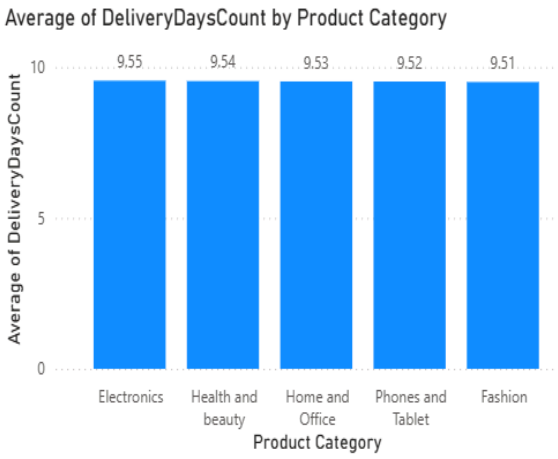
10. Wait Times Correlated with Demographics and Care: Explore how average wait times vary across different product categories to optimize scheduling and staffing.

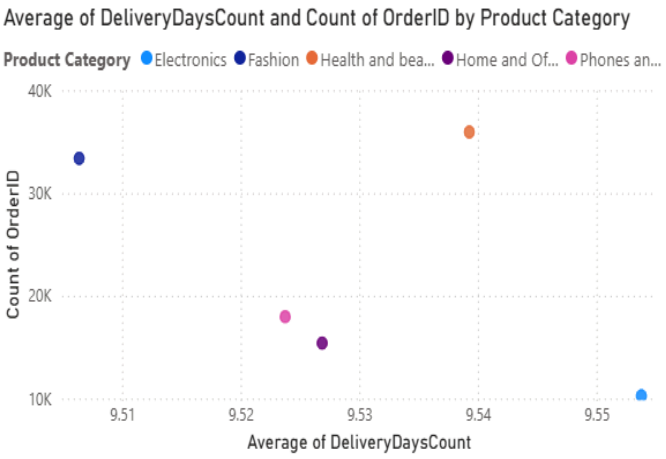
A. Approach

To analyze wait times, I calculated the delivery duration by finding the difference between the order date and delivery date. Using this metric, I compared the **average delivery wait time across different product categories** to understand how delivery performance varies. I also analyzed order volumes alongside wait times to identify which categories require operational attention for better scheduling and staffing.

Visualization

- **Average Delivery Days by Product Category (Bar Chart)**
  This visual shows the average wait time for each product category, making it easy to compare delivery performance across categories.
- **Average Delivery Days vs Order Volume (Scatter Plot)**
  This visual plots average wait time against the number of orders for each product category, helping identify categories with both high demand and longer delivery durations.





Insights

- The average delivery time across all product categories is fairly consistent, ranging between **9.5 and 9.6 days**, indicating a standardized delivery process.
- Categories like **Electronics and Health & Beauty** show slightly higher average wait times compared to others.

- **Fashion and Electronics** have high order volumes, meaning even small delays in these categories can significantly impact customer satisfaction.
- Categories with lower order volumes but similar wait times indicate opportunities to optimize resources without affecting service levels.

Recommendations

- **Prioritize staffing and logistics resources** for high-volume categories such as Fashion and Electronics to reduce the impact of delivery delays.
- **Optimize scheduling during peak demand periods** for categories with consistently higher order volumes to prevent bottlenecks.
- **Investigate category-specific fulfillment processes** to identify minor inefficiencies causing slightly higher wait times.
- **Continuously monitor average delivery days by category** using dashboards to proactively adjust staffing and delivery capacity.

11. Explore if there is any relationship between the Delivery type and waiting time between ordering and receiving an item.

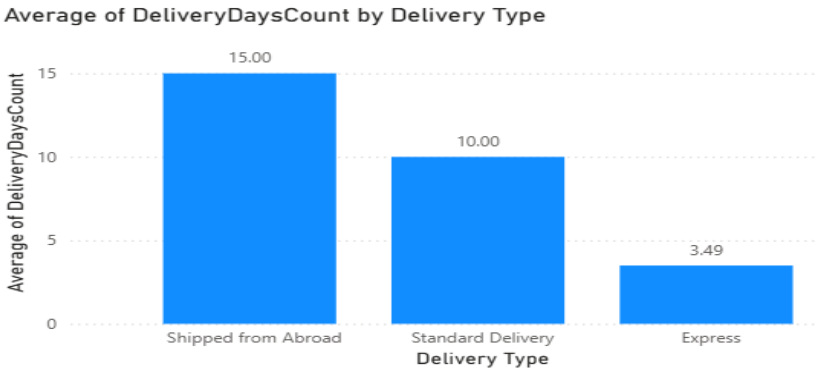
A. **Approach**

To understand the relationship between delivery type and waiting time, I calculated the delivery duration by measuring the difference between the order date and delivery date. I then compared the **average waiting time across different delivery types** to identify how delivery method impacts the time taken to receive an item.

Visualization

A **bar chart** was used to display the **average delivery days by delivery type**.

This visual clearly compares waiting times for *Shipped from Abroad*, *Standard Delivery*, and *Express Delivery*, making it easy to identify differences in delivery performance.



Insights

- **Shipped from Abroad** has the highest average waiting time at around **15 days**, indicating longer transit and customs-related delays.
- **Standard Delivery** shows a moderate waiting time of approximately **10 days**, representing regular domestic fulfillment timelines.
- **Express Delivery** has the lowest waiting time at around **3.5 days**, confirming its effectiveness for faster order fulfillment.
- The clear variation across delivery types shows a **strong relationship between delivery method and waiting time**.

Recommendations

- **Promote Express Delivery for time-sensitive customers**, especially for high-value or urgent orders.
- **Set clear delivery expectations** for "Shipped from Abroad" orders to manage customer satisfaction.
- **Optimize logistics for standard delivery** to reduce waiting time and narrow the gap with express services.
- **Use delivery-type insights to plan staffing and courier partnerships**, ensuring adequate resources for faster delivery options.

12.Is there any relationship between shipping charges and product type?

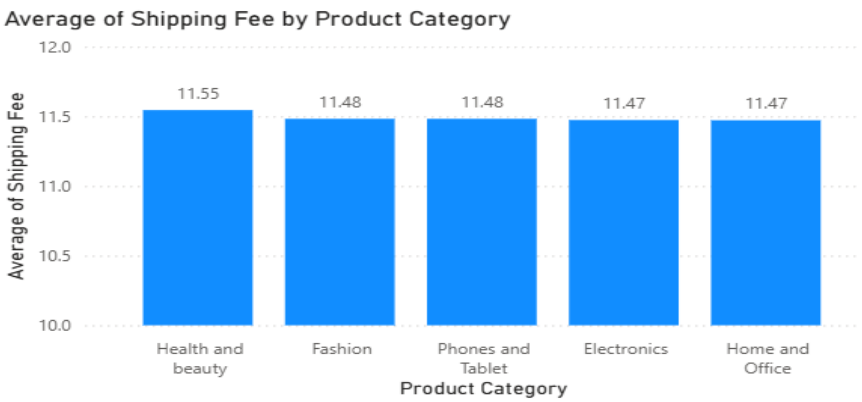
A. Approach

To analyze whether shipping charges vary by product type, I calculated the **average shipping fee for each product category** and compared them across categories. This helps identify whether product type meaningfully influences shipping costs or if charges remain largely uniform.

Visualization

A **bar chart** showing the **average shipping fee by product category** was used.

This visualization allows quick comparison of shipping costs across categories such as Health & Beauty, Fashion, Electronics, Phones & Tablets, and Home & Office.



Insights

- **Shipping charges are almost uniform across all product categories**, with averages tightly clustered around **₹11.45–₹11.55**.
- **Health & Beauty** has a marginally higher average shipping fee, but the difference is **not significant**.
- There is **no strong evidence** that product type materially impacts shipping charges in this dataset.
- This suggests shipping fees are likely **standardized** rather than weight-, size-, or category-driven.

Recommendations

- **Avoid category-based shipping pricing**, as customers are not experiencing meaningful cost differences by product type.
- Consider introducing **weight-based or distance-based shipping fees** if cost optimization is a business goal.
- Use **flat-rate shipping as a marketing advantage**, since pricing consistency improves customer trust.
- If logistics costs differ internally, **re-evaluate cost absorption strategies** to prevent margin leakage.

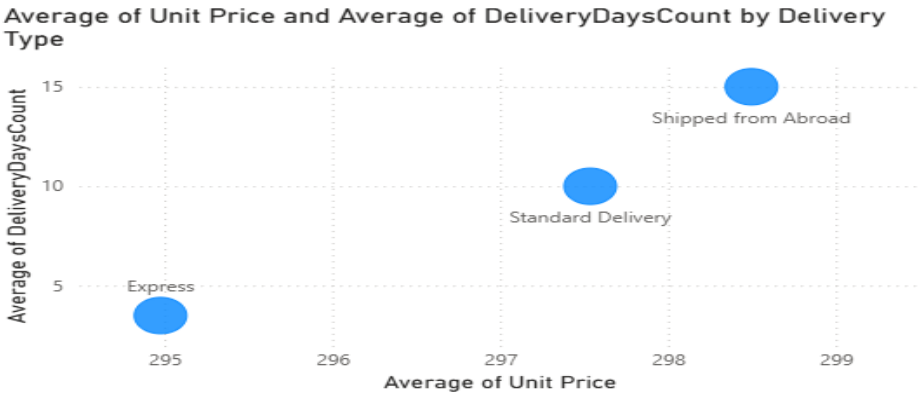
13. Come up with strategies to decrease the low rating orders after analyzing different factors like waiting time, shipping type, unit price, etc.

A. Approach

To reduce low-rated orders, I analyzed the relationship between **delivery type, waiting time, and unit price** and how these operational factors influence customer experience. The goal is to identify friction points that lead to dissatisfaction and design targeted corrective strategies.

Visualization

- A **bubble/scatter chart** was used with:
  - **X-axis:** Average Unit Price
  - **Y-axis:** Average Delivery Days
  - **Bubble size / category:** Delivery Type
- This helps compare how **price and delivery speed together** vary across delivery types and potentially impact customer ratings.



Insights

- **Shipped from Abroad** has the **highest delivery time (~15 days)** and **highest unit price**, making it the **highest-risk segment for low ratings**.
- **Standard Delivery** sits in the middle on both price and wait time, suggesting **moderate dissatisfaction risk**.
- **Express Delivery** has the **lowest wait time (~3–4 days)** and relatively lower unit price, indicating **better customer satisfaction potential**.
- Longer wait times combined with higher prices **amplify negative customer perception**, leading to low ratings.

Recommendations

- **Set clear delivery expectations** for "Shipped from Abroad" orders (explicit ETA warnings before checkout).
- Offer **price-based compensation** (discounts, vouchers, or free returns) for long-wait orders.
- **Incentivize Express Delivery** for high-value orders to reduce dissatisfaction.
- Introduce **proactive order tracking and status updates** to reduce anxiety during long waits.
- Monitor **ratings by delivery type** as a KPI and trigger interventions when thresholds drop.

14. Using the time intelligence DAX function, create a table to compare each month's sales with the previous year's same month's total sales. So there will be four columns in the output year, month, total sales, previous_years_sales.

A. **Approach**

Time intelligence DAX functions were used to compare **current month sales with the same month from the previous year**. First, total monthly sales were calculated using date hierarchy (Year–Month). Then, a **previous year sales measure** was created using time intelligence to shift the date context by one year. This enables **like-to-like YoY comparison**, removing seasonality bias and making performance trends easy to evaluate.

Visualization

A **table visual** was used with the following columns:

Year , Month , Current Year Sales , Previous Year Sales

This format allows **direct row-level comparison** for each month across years and helps quickly identify growth or decline patterns.

Year	Month	Current Year Sales	Previous_Year_Sales
2019	January	1432370	1410978
2019	February	1150174	1190825
2019	March	1415226	1328083
2019	April	1343959	1333065
2019	May	1435070	1310744
2019	June	1413734	1376271
2019	July	1536245	1369416
2019	August	1451249	1455698
2019	September	1403828	1348170
2019	October	1500747	1356434
2019	November	1393986	1406514
2019	December	1293016	1398984
2020	January	3009297	1432370
2020	February	2455413	1150174
2020	March	3030981	1415226
2020	April	2833880	1343959
2020	May	2859603	1435070
2020	June	1268901	1413734
2020	July	1437864	1536245
2020	August	1372313	1451249
Total		**107239212**	**83262658**

Insights

- **2020 shows a strong YoY growth** compared to 2019, especially from **January to May**, where sales more than doubled.
- Certain months (e.g., **June–August 2020**) show **slower growth or decline** compared to the previous year, indicating possible seasonality or operational constraints.
- The **total current year sales (10.7M)** significantly exceed **previous year sales (8.3M)**, confirming overall business growth.
- Month-level comparison highlights **inconsistent growth**, meaning performance is not uniform across the year.

Recommendations

- **Double down on high-growth months** (early 2020) with increased inventory and marketing spend.
- Investigate **underperforming months** to identify causes such as supply issues, demand drops, or external factors.
- Use YoY comparison as a **standard KPI** for monthly performance reviews.
- Combine this analysis with **category-level or region-level YoY trends** for more granular decision-making.

15. What do you understand about the PowerBI gateway? What are its use cases?

A. Power BI Gateway acts as a **secure link between on-premises data sources and the cloud Power BI service**. It allows organizations to **refresh and update reports automatically** without manually moving data to the cloud.

Key Use Cases:

- **On-Premises Data Access:** Connects local databases like SQL Server, Excel, or ERP/CRM systems to Power BI for seamless reporting.
- **Automated Data Refresh:** Ensures dashboards stay updated with the latest data from local systems without manual intervention.
- **Real-Time Insights:** Supports near real-time analytics by bridging local business systems with Power BI.
- **Hybrid Environments:** Enables organizations to work with both cloud and on-premises data sources together.
- **Secure and Compliant Data Transfer:** Maintains data security and compliance while moving information between local systems and the cloud.

16. How would you approach this problem, if the objective and subjective questions weren't given?

A. If the objective and subjective questions are not provided, I would start by understanding the dataset thoroughly and identifying patterns or areas that can provide actionable business insights. My approach would focus on exploring, analyzing, and deriving meaningful insights to guide decision-making.

1. Data Exploration and Cleaning

First, I would thoroughly explore the dataset to understand all tables, columns, and relationships. I would check for missing values, inconsistencies, or duplicate data and clean it to ensure accuracy. This step lays a strong foundation for meaningful analysis.

2. Define Key Metrics and KPIs

Next, I would identify the key metrics that matter most to the business, such as total sales, revenue growth, delivery efficiency, customer satisfaction, or product performance. Defining these KPIs helps focus the analysis on areas that can create real business impact.

3. Perform Exploratory Analysis

I would then analyze trends, patterns, and anomalies in the data. This includes looking at seasonal variations, sudden spikes or drops, and other unusual behaviors to uncover insights that might not be immediately obvious.

4. Prioritize and Visualize Insights

After exploring the data, I would prioritize insights based on their potential value to the business. Then I would create clear, interactive dashboards that allow stakeholders to drill down by region, product, or time, helping them make informed decisions quickly.

5. Generate Hypotheses

I would develop possible explanations for observed trends or anomalies, forming hypotheses that can guide deeper analysis. This step helps to focus investigations on the most meaningful patterns rather than random noise.

6. Provide Actionable Recommendations

Finally, I would translate the insights into practical recommendations for the business. For example, suggesting ways to improve delivery times, optimize inventory, boost sales in underperforming regions, or enhance customer experience.